

1. Lab5Part1

a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. First string 2. Second string	1. Display first string substring 2. Display second string
Processing Required	Solution Alternative
<pre> Read firstString if (firstString.length() >= 10) Print firstString.substring(4,10) else Print "String length less than 10" Read secondString if (firstString.contains(secondString)) { Print "Second String exists in First String" Replace secondString in firstString with "Selamat Datang" Print firstString } else Print "Second String not exists in First String" </pre>	1. Define first and second strings as constants. 2. Define first and second strings as input values.

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
First String Second String	<pre> 1. Enter firstString 2. if (firstString.length() >= 10) Print firstString.substring(4,10) else Print "String length less than 10" 3. Enter secondString 4. if (firstString.contains(secondString)) { Print "Second String exists in First String" Replace secondString in firstString with "Selamat Datang" Print firstString } else Print "Second String not exists in First String" 5. End </pre>	-	1. First String 2. Second String

c) Flowchart



d) Pseudocode

Start

```
Read firstString
if (firstString.length() >= 10)
    Print "Substring from 5 to 10 " + firstString.substring(4,10) + ", length " +
    firstString.length()
else
    Print "String length less than 10"
Read secondString
if (firstString.contains(secondString)) {
    Print "Second String exists in First String"
    Replace secondString in firstString with "Selamat Datang"
    Print "New Statement in String 1 : " + firstString
} else
    Print "Second String not exists in First String"
```

End

e) Java code

```
import java.util.Scanner;

public class Lab5Part1{
    // Ngo Yuk Hang, 228991, Lab 5 Part 1
    public static void main(String[] args){
        // Init
        Scanner scanner = new Scanner(System.in);

        // First String
        System.out.println("Enter first String : ");
        String firstString = scanner.nextLine();
        int firstStringLength = firstString.length();
        if(firstStringLength >= 10){
            System.out.println("Substring from 5 to 10 " + firstString.substring(4,10) + ", length " + firstStringLength);
        } else {
            System.out.println("String length less than 10");
        }

        // Second String
        System.out.println("\nEnter second String : ");
        String secondString = scanner.nextLine();
        if(firstString.contains(secondString)){
            System.out.println("Second String exists in First String");
            int indexInFirstString = firstString.indexOf(secondString);
            String newFirstString = firstString.substring(0,indexInFirstString) + "Selamat Datang" + firstString.substring(indexInFirstString + secondString.length());
            System.out.println("\nNew Statement in String 1 : " + newFirstString);
        } else {
            System.out.println("Second String not exists in First String");
        }
    }
}
```

2. Lab5Part2

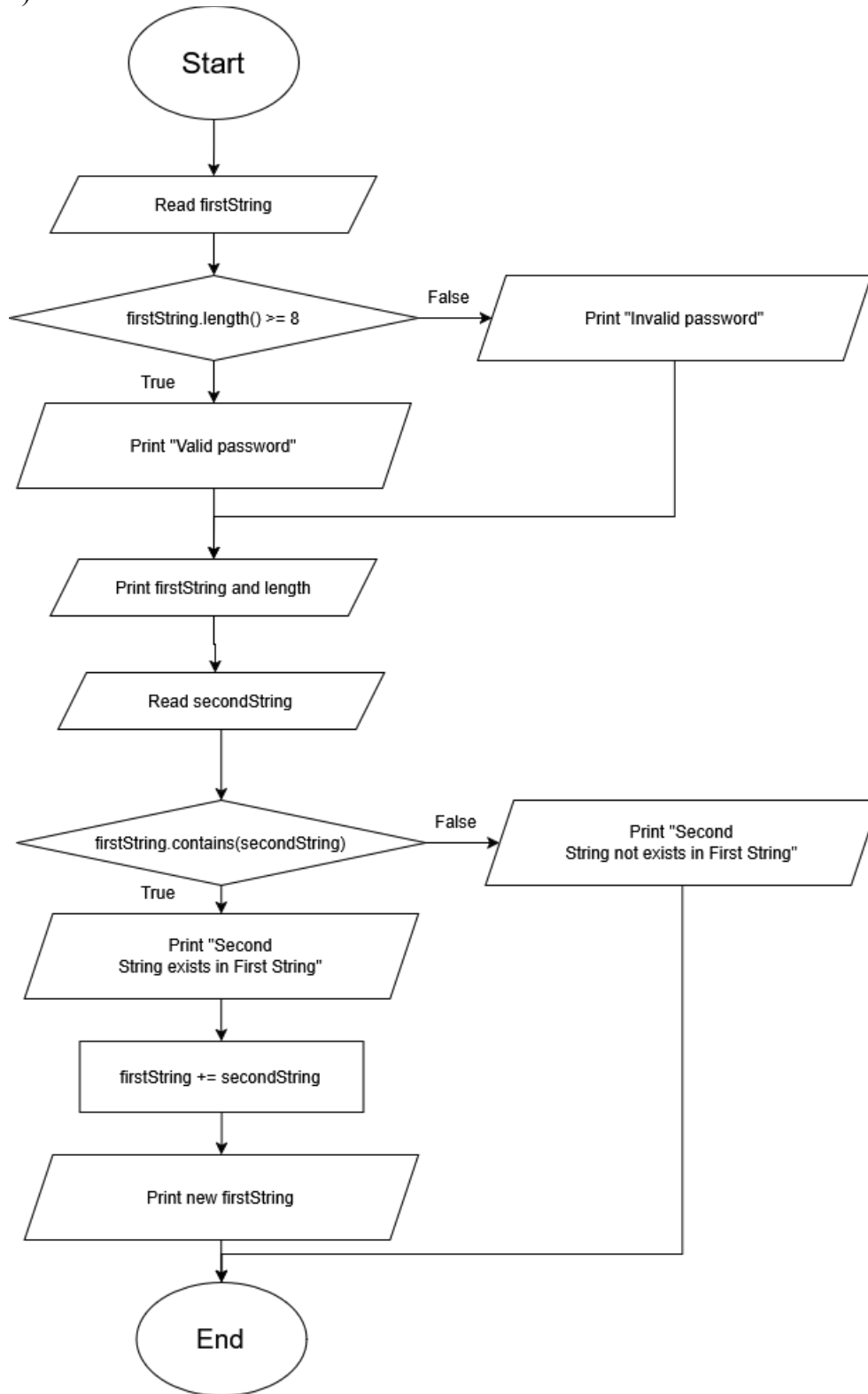
a) Problem Analysis Chart (PAC)

Given Data	Required Results
1. First String 2. Second String	1. Validity of First String 2. New First String
Processing Required	Solution Alternative
<pre> Read firstString if (firstString.length() >= 8) Print "Valid password" else Print "Invalid password" Read secondString if (firstString.contains(secondString)) Print "Second String exists in First String" else{ Print "Second String not exists in First String" firstString += secondString } </pre>	<pre> 1. Define first and second strings as constants. 2. Define first and second strings as input values. </pre>

b) Input Process Output (IPO)

Input	Processing	Module Reference	Output
First String Second String	<pre> 1. Read firstString 2. if (firstString.length() >= 8) Print "Valid password" else Print "Invalid password" 3. Read secondString 4. if (firstString.contains(secondString)) Print "Second String exists in First String" else{ Print "Second String not exists in First String" firstString += secondString } 5. End </pre>	-	1. Validity of First String 2. New First String

c) Flowchart



d) Pseudocode

Start

```
Read firstString
if (firstString.length() >= 8)
    Print "Valid password"
else
    Print "Invalid password"
Print firstString and length
Read secondString
if (firstString.contains(secondString)) {
    Print "Second String exists in First String"
} else {
    Print "Second String not exists in First String"
    firstString += secondString
    Print firstString
}
```

End

e) Java code

```
import java.util.Scanner;

public class Lab5Part2{
    // Ngo Yuk Hang, 228991, Lab 5 Part 2
    public static void main(String[] args){
        // Init
        Scanner scanner = new Scanner(System.in);

        // First String
        System.out.println("Enter first String : ");
        String firstString = scanner.nextLine();
        int firstStringLength = firstString.length();
        if (firstStringLength >= 8)
            System.out.println("Valid password");
        else
            System.out.println("Invalid password");
        System.out.println("First String : \" + firstString + "\" with length " + firstStringLength);

        // Second String
        System.out.println("\nEnter second String : ");
        String secondString = scanner.nextLine();
        if (firstString.contains(secondString)){
            System.out.println("Second String exists in First String");
        } else {
            System.out.println("Second String not exists in First String");
            firstString += secondString;
            System.out.println("\nNew Statement in String 1 : " + firstString);
        }
    }
}
```